

Enes Senel

Senior Scientist · Machine Learning & Computational Biology

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PROFILE

Senior Scientist and computer scientist with 10+ years building machine-learning systems across biotech, pharma, and medical robotics. Track record of leading multidisciplinary projects end-to-end — from problem framing and method design to production pipelines, open-source releases, and regulatory-grade documentation. Currently leading multi-omics strategy at Johnson & Johnson Innovative Medicine; previously shipped real-time computer vision for image-guided robotics and deep-learning systems for pharmaceutical QC. Comfortable as technical lead, individual contributor, or bridge between scientific, engineering, and clinical stakeholders.

CORE SKILLS

Leadership: Technical leadership, scientific strategy, mentorship, stakeholder management, cross-functional collaboration

Machine Learning: Deep learning, computer vision, optimal transport, transformers / language models (ProtBERT), domain adaptation

Engineering: Python, R, C++, CUDA, PyTorch, TensorFlow, Nextflow, Docker, SQL, reproducible production pipelines

Domains: Computational biology, single-cell & spatial omics, medical robotics, image-guided systems, pharma QC, neuroscience

SELECTED EXPERIENCE

Senior Scientist, Computational Biology

Johnson & Johnson Innovative Medicine · Berlin, Germany · Aug 2025 — Present

- Technical lead for single-cell multi-omics in oncology & immunology — sets analytical strategy across parallel programs.
- Mentors junior scientists, manages external consultants, and aligns computational deliverables with clinical and discovery stakeholders.
- Translates ML outputs into go/no-go decisions on assets in development.

Data Scientist, Computational Biology

Johnson & Johnson Innovative Medicine · Berlin, Germany · Oct 2023 — Jul 2025

- Architected Nextflow/Python production pipelines for scRNA-seq, CITE-seq, and multi-modal analyses — turned week-long manual workflows into reproducible runs.
- Led single-cell analyses underpinning biomarker discovery for a clinical-stage cell therapy (CARTITUDE-1) and CMC process characterization.
- Established team conventions for reproducibility (containerization, QC standards) adopted as the default for new projects.

Doctoral Researcher — Computational Biology & ML

Max Delbrück Center for Molecular Medicine · Berlin, Germany · Oct 2019 — Sep 2023

- Designed Optocoder, an ML pipeline for decoding barcoded transcripts — published in NAR Genomics & Bioinformatics, adopted by external labs.
- Re-architected novoSpaRc (optimal-transport tissue reconstruction) for scale; co-author on Nature Protocols paper.
- Co-inventor on US patent US20240257914A1 for 3D spatial gene-expression reconstruction.

ML Research Engineer · Computer Vision & Robotics (combined)

Coriolis Pharma · Max Planck Institute · Ozyegin Robotics Lab · Munich · Frankfurt · Istanbul · 2014 — 2019

- **Coriolis Pharma (2019):** Shipped end-to-end deep-learning system (PyTorch/TensorFlow) for automated particle detection in pharmaceutical QC; handed off as a self-serve internal tool.
- **Max Planck (2018–2019):** Multi-omics ML integration bridging proteomic and transcriptomic datasets — early work in cross-modality reasoning.
- **Ozyegin Robotics Lab (2014–2016):** Real-time needle-tip localization and tracking from ultrasound for image-guided biopsy robots; C++/CUDA software coupling vision with robot control. 4 peer-reviewed publications (ICRA, ICAR, Mechatronics).

EDUCATION

Ph.D. Computational Biology — Humboldt University of Berlin · Max Delbrück Center · 2019 — 2026 (*exp.*)

M.Sc. Computer Science — Technical University of Munich · 2016 — 2019 · DAAD MSc Scholarship

B.Sc. Computer Science — Ozyegin University, Istanbul · 2012 — 2016

HIGHLIGHTS

Patent: US20240257914A1 — Method and system for 3D reconstruction of tissue gene expression data (2022).

Open source: [Optocoder](#) · [novoSpaRc](#) — ML tools used by labs and industry for spatial transcriptomics.

Selected publications: Nature Protocols (2021), NAR Genomics & Bioinformatics (2022), Blood Cancer Discovery (CARTITUDE-1, 2025, under revision), Cytotherapy (2021), Mechatronics (2019), ICRA (2016, 2017).

Service: Helmholtz Juniors (elected member, ~8,000 PhD students, 2022–2023); MDC PhD Representatives (2020–2023).

LANGUAGES

English (Fluent) · German (Advanced) · Turkish (Native)